

9(a)(i)	<i>absorption spectrum</i> absorption of different wavelengths (of light) by (named) pigment(s) ; <i>action spectrum</i> rate of photosynthesis at different wavelengths (of light) ;	2
9(a)(ii)	any three from : 1 chlorophyll b / carotenoids / accessory pigments (are present) ; 2 (these other pigments) absorb light (energy), between 450 and 550 nm / of wavelengths that chlorophyll a does not absorb ; 3 energy passed on to, reaction centre / chlorophyll a / primary pigment ; 4 for, light-dependent reactions / photophosphorylation / photolysis (of water) / photoactivation ;	3

Question	Answer	Marks																				
9(b)(i)	<i>any four from :</i> 1 increase then decrease ; 2 optimum temperature at 29 °C ; 3 data quote – two temperatures and two rates with units ; <table><tr><th>temperature / °C ±0.5</th><th>rate of photosynthesis / mg g⁻¹ h⁻¹ ±0.05</th></tr><tr><td>2.5</td><td>0.75</td></tr><tr><td>5</td><td>1.1</td></tr><tr><td>10</td><td>1.8</td></tr><tr><td>20</td><td>2.95</td></tr><tr><td>29</td><td>3.5</td></tr><tr><td>30</td><td>3.45</td></tr><tr><td>40</td><td>2.65</td></tr><tr><td>45</td><td>1.5</td></tr><tr><td>48</td><td>0.5</td></tr></table> <i>increase / before 29 °C</i> 4 more kinetic energy for, enzyme (activity) / AW ; 5 temperature is the limiting factor ; <i>decrease / after 29 °C</i> 6 denaturation of, rubisco / enzymes / oxygen-evolving complex ; 7 AVP ; e.g. photorespiration limiting the rate at above 25 °C	temperature / °C ±0.5	rate of photosynthesis / mg g ⁻¹ h ⁻¹ ±0.05	2.5	0.75	5	1.1	10	1.8	20	2.95	29	3.5	30	3.45	40	2.65	45	1.5	48	0.5	4
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Question	Answer	Marks
9(b)(ii)	3.525 to 3.675 ; mg g ⁻¹ h ⁻¹ ;	2

Question	Answer	Marks
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